HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **PLC programming** for new automation lines, integrating **Rockwell** and **Siemens S7** controllers to manage robotic arms and vision systems, reducing manual assembly steps by **45%**.
- Managed **production equipment troubleshooting** for a live manufacturing environment, diagnosing and resolving complex automation issues on **servo drives** and **motion control systems**, ensuring minimal downtime.
- Developed **GAMP documentation** (URS, FDS, FAT, SAT) in collaboration with project teams for new CapEx automation installations, ensuring compliance with pharmaceutical manufacturing standards.
- Analyzed opportunities for process improvement, defining and selecting appropriate **automation technologies** and hardware for department upgrades, contributing to a **15%** increase in overall equipment effectiveness.
- Provided on-call support for **automation issues** during out-of-hours shifts, interrogating **PLC code** and network protocols to quickly identify and rectify system faults.
- Coached area technicians and junior control engineers on **automation network protocols** such as **EtherNet IP** and **Modbus**, enhancing team's problem-solving capabilities for control system issues.
- Containerized existing Java Spring Boot services using **Docker** and deployed them to an **AWS EKS** cluster, automating CI/CD pipelines with **Jenkins** for streamlined application updates.

SKILLS

Automation & Control – PLC, SCADA, DCS, HMI, Robotics, Vision Systems, Servo Motors, Motion Control, GAMP, URS, FDS, FAT, SAT, Machinery Directive, CE Compliance, EN60204-1

PLC Programming & Vendors – Rockwell, Siemens S7, Telemecanique

Networking & Protocols – EtherNet IP, DeviceNet, ControlNet, Modbus, Profibus

Software & Tools – Python, Java, Spring Boot, SQL, Docker, Jenkins, Jira, Confluence, Splunk, Dynatrace, AWS (EC2, S3, Lambda, CloudWatch)

Manufacturing Systems – MES Integration, Change Management, Capital Project Justification, Budget Management, EHS Incident Investigation, RCA

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy, scikit-learn

Engineering Degree & Experience – Bachelor of Engineering, MSc Data Analytics, 3.5 Years Experience, GMP Environment, Quality Assurance

PROJECTS

Automated Pharmaceutical Packaging Line Control System

- Designed and programmed the control logic for a high-speed pharmaceutical packaging line using **Siemens S7 PLCs**, integrating **vision systems** for quality inspection and robotic pick-and-place mechanisms.
- Developed comprehensive **FAT and SAT** documentation for the packaging line, including detailed test cases and validation reports, ensuring adherence to **GMP** and EU Machinery Directives.
- Implemented fault-finding routines within the **PLC code** to diagnose and resolve issues related to servo motor synchronization and conveyor belt speed control, reducing troubleshooting time by **30%**.

Java-based Manufacturing Data Aggregation Service

- Engineered a **Java Spring Boot** microservice to collect real-time data from various manufacturing floor equipment via **Modbus** and **EtherNet IP** protocols, processing approximately **5,000 events per minute**.
- Developed REST APIs for the service to expose aggregated data to the site's **MES**, facilitating improved process monitoring and operational decision-making.
- Containerized the aggregation service using **Docker** and deployed it on an **AWS EC2** instance, managed through a **Jenkins** CI/CD pipeline for automated builds and deployments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)